

Class 12: RNASeq with DESeq2

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Backgorund

Today we will analyze some RNASeq data form Himes et al. on the effects of a common steroid (dexamethasone, also called “dex”) on airway smooth muscle cells (ASMs).

For this analysis we need two main inputs

- **countData**: a table of **counts** per gene (in rows) across experiments (in columns)
- **colData**: **metadata** about the design of the experiments. The rows match the columns in countData.

Data Import

```
counts <- read.csv("airway_scaledcounts.csv", row.names = 1)
metadata <- read.csv("airway_metadata.csv")
```

Let's have a wee peak at our **counts** data.

```
head(counts)
```

	SRR1039508	SRR1039509	SRR1039512	SRR1039513	SRR1039516
ENSG000000000003	723	486	904	445	1170
ENSG000000000005	0	0	0	0	0
ENSG000000000419	467	523	616	371	582
ENSG000000000457	347	258	364	237	318
ENSG000000000460	96	81	73	66	118
ENSG000000000938	0	0	1	0	2

	SRR1039517	SRR1039520	SRR1039521
ENSG000000000003	1097	806	604
ENSG000000000005	0	0	0
ENSG000000000419	781	417	509
ENSG000000000457	447	330	324
ENSG000000000460	94	102	74
ENSG000000000938	0	0	0

and the metadata

```
metadata
```

	id	dex	celltype	geo_id
1	SRR1039508	control	N61311	GSM1275862
2	SRR1039509	treated	N61311	GSM1275863
3	SRR1039512	control	N052611	GSM1275866
4	SRR1039513	treated	N052611	GSM1275867
5	SRR1039516	control	N080611	GSM1275870
6	SRR1039517	treated	N080611	GSM1275871
7	SRR1039520	control	N061011	GSM1275874
8	SRR1039521	treated	N061011	GSM1275875

Q1. How many “genes” are in this dataset?

```
nrow(counts)
```

```
[1] 38694
```

Q2. How many experiments (i.e. columns in `counts` or rows in `metadata`) are there?

```
ncol(counts)
```

```
[1] 8
```

Q3. How many “control” experiments are there in the dataset?

```
sum(metadata$dex=="control")
```

```
[1] 4
```

Toy analysis example

1. Extract the “control” columns from `counts`
2. Calculate the mean value for each gene in these “control” columns 3-4. Do the same for the “treated” columns.
3. Compare these mean values for each gene.

Step 1.

```
control.inds <- metadata$dex=="control"  
control.counts <- counts[,control.inds]
```

Step 2.

```
control.mean <- rowMeans(control.counts)
```

Step 3 and 4.

```
treat.inds <- metadata$dex=="treated"  
treat.counts <- counts[,treat.inds]  
treat.mean <- rowMeans(treat.counts)
```

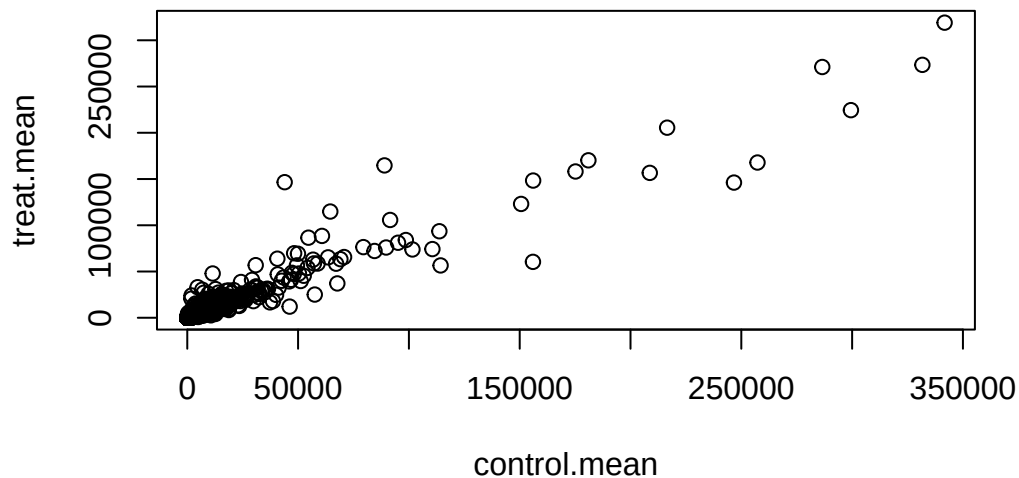
For ease of book-keeping we can store these together in one data frame called `meancounts`

```
meancounts <- data.frame(control.mean, treat.mean)  
head(meancounts)
```

	control.mean	treat.mean
ENSG000000000003	900.75	658.00
ENSG000000000005	0.00	0.00
ENSG0000000000419	520.50	546.00
ENSG0000000000457	339.75	316.50
ENSG0000000000460	97.25	78.75
ENSG0000000000938	0.75	0.00

Plot these against each other

```
plot(meancounts)
```

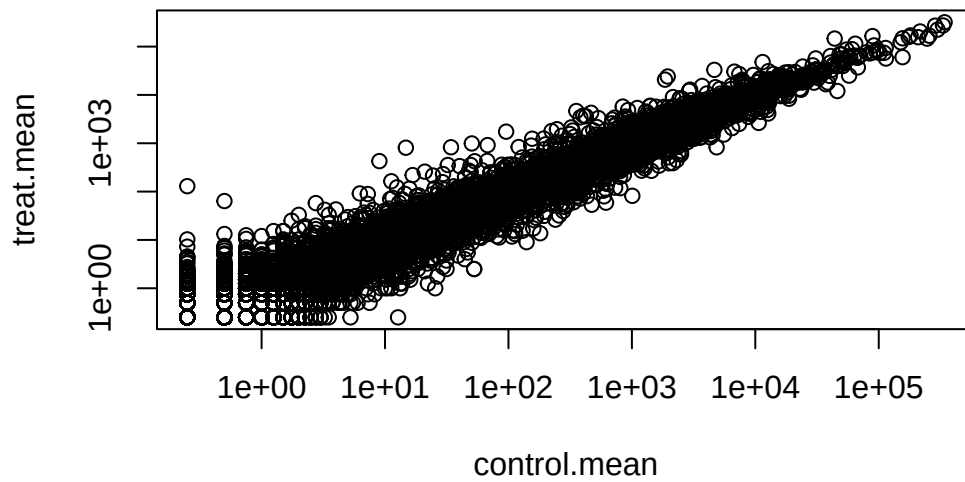


This is screaming at me to log transform this data

```
plot(meancounts, log="xy")
```

Warning in xy.coords(x, y, xlabel, ylabel, log): 15032 x values <= 0 omitted from logarithmic plot

Warning in xy.coords(x, y, xlabel, ylabel, log): 15281 y values <= 0 omitted from logarithmic plot



We use log2 “fold-change” as a way to compare

```
# treated/control  
log2(10/10)
```

```
[1] 0
```

```
log2(20/10)
```

```
[1] 1
```

```
log2(10/20)
```

```
[1] -1
```

```
log2(40/10)
```

```
[1] 2
```

```

meancounts$log2fc <- log2(meancounts$treat.mean/
                           meancounts$control.mean)
head(meancounts)

```

	control.mean	treat.mean	log2fc
ENSG000000000003	900.75	658.00	-0.45303916
ENSG000000000005	0.00	0.00	NaN
ENSG000000000419	520.50	546.00	0.06900279
ENSG000000000457	339.75	316.50	-0.10226805
ENSG000000000460	97.25	78.75	-0.30441833
ENSG000000000938	0.75	0.00	-Inf

A common “rule-of-thumb” threshold for calling something “up” regulated is a log2-fold-change of +2 or greater. For down regulated -2 or less.

Q How many genes are “up” regulated at the +2 log2FC threshold?

```

table(meancounts$log2fc>=2)

```

```

FALSE  TRUE
23348  1910

```

Filter out zero

```

x <- c(1,5,0,5)
which(x==0)

```

```
[1] 3
```

```

y <- data.frame(a=c(1,5,0,5), b=c(1,0,5,5))
y

```

```

  a b
1 1 1
2 5 0
3 0 5
4 5 5

```

```
which(y==0, arr.ind = TRUE)
```

```
      row col
[1,]   3   1
[2,]   2   2
```

```
zero.inds <- which(meancounts[,1:2]==0, arr.ind = TRUE)[,1]
mygenes <- meancounts[-zero.inds,]
```

```
sum(mygenes$log2fc>=2)
```

```
[1] 314
```

How many genes are “down” regulated (at the -2 log2FC threshold)?

```
table(meancounts$log2fc>=-2)
```

```
FALSE  TRUE
 2212 23046
```

DESeq analysis

Let's do this with DESeq2 and put some stats behind these numbers.

```
library(DESeq2)
```

DESeq wants 3 things for analysis: countData, colData and design.

```
dds <- DESeqDataSetFromMatrix(countData = counts,
                              colData = metadata,
                              design = ~dex)
```

converting counts to integer mode

Warning in DESeqDataSet(se, design = design, ignoreRank): some variables in design formula are characters, converting to factors

The main function in the DESeq package to run analysis is called DESeq().

```
dds <- DESeq(dds)
```

estimating size factors

estimating dispersions

gene-wise dispersion estimates

mean-dispersion relationship

final dispersion estimates

fitting model and testing

Get the results out of this DESeq object with the function `results()`

```
res <- results(dds)
head(res)
```

log2 fold change (MLE): dex treated vs control

Wald test p-value: dex treated vs control

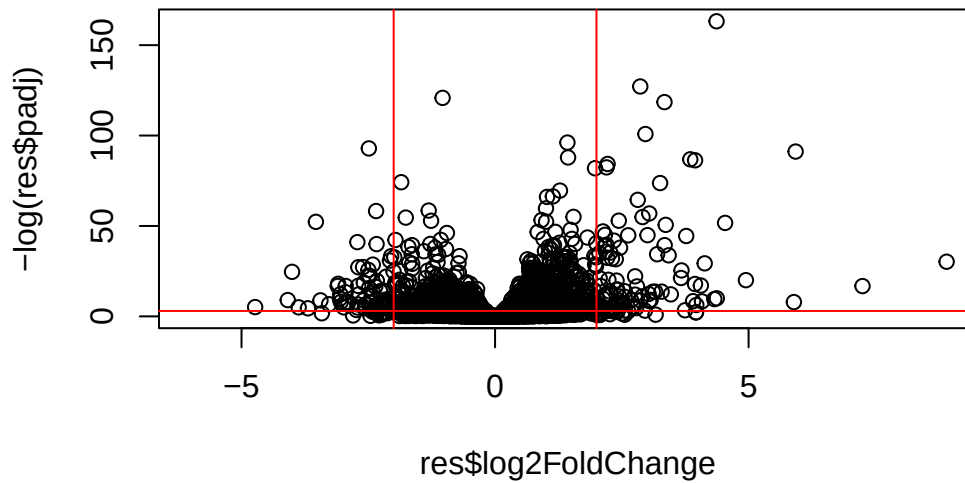
DataFrame with 6 rows and 6 columns

	baseMean	log2FoldChange	lfcSE	stat	pvalue
	<numeric>	<numeric>	<numeric>	<numeric>	<numeric>
ENSG000000000003	747.194195	-0.350703	0.168242	-2.084514	0.0371134
ENSG000000000005	0.000000	NA	NA	NA	NA
ENSG000000000419	520.134160	0.206107	0.101042	2.039828	0.0413675
ENSG000000000457	322.664844	0.024527	0.145134	0.168996	0.8658000
ENSG000000000460	87.682625	-0.147143	0.256995	-0.572550	0.5669497
ENSG000000000938	0.319167	-1.732289	3.493601	-0.495846	0.6200029
	padj				
	<numeric>				
ENSG000000000003	0.163017				
ENSG000000000005	NA				
ENSG000000000419	0.175937				
ENSG000000000457	0.961682				
ENSG000000000460	0.815805				
ENSG000000000938	NA				

Volcano Plot

This is a plot of log2FC vs adjusted p-value

```
plot(res$log2FoldChange, -log(res$padj))
abline(v=c(-2,2), col="red")
abline(h=-log(0.05), col="red")
```



Save our results

```
write.csv(res, file = "myresults.csv")
```

Make prettier volcano plot

```
library(ggplot2)

mycols <- rep("grey", nrow(res))
mycols[abs(res$log2FoldChange)>2] <- "blue"
mycols[res$padj>=0.05] <- "red"
```

```
ggplot(res) +  
  aes(log2FoldChange, -log(padj))+  
  geom_point(col=mycols)
```

Warning: Removed 23549 rows containing missing values or values outside the scale range (`geom\_point()`).

